

MOSKOVA DOMODEDOVO, Russian Federation

WMO: 370008

Lat: 55.409N

Lon: 37.906E

Elev: 179

StdP: 99.19

Time Zone: 3.00 (E03)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.1	(c) -21.0	(d) -27.0	(e) 0.3	(f) -23.9	(g) -23.9	(h) 0.4	(i) -21.0	(j) 9.9	(k) -2.7	(l) 9.1	(m) -3.0	(n) 2.0	(o) 40	(p) 0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	30.2	19.5	28.2	18.8	26.2	18.1	21.0	26.9	20.1	25.9	19.3	24.8	3.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.2	23.2	18.1	13.3	22.2	17.2	12.5	21.4	61.5	27.0	58.4	26.0	55.7	24.4	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	8.2	7.3	-27.1	32.0	3.8	2.1	-29.8	33.5	-32.0	34.7	-34.1	35.9	-36.9	37.5
			-27.3	22.6	3.7	1.2	-30.0	23.5	-32.2	24.2	-34.3	24.9	-37.0	25.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-7.7	-7.1	-1.7	6.6	13.0	16.3	19.6	17.6	12.0	5.7	-0.3	-4.3
	DBStd	10.68	6.97	6.93	5.05	4.69	4.70	3.84	3.34	3.60	3.66	4.39	5.02	5.95
	HDD10.0	2457	549	478	363	123	24	2	0	0	21	144	308	444
	HDD18.3	4673	807	711	622	352	173	82	25	57	192	390	558	703
	CDD10.0	956	0	0	0	21	118	193	297	236	79	12	0	0
	CDD18.3	130	0	0	0	0	8	23	63	34	2	0	0	0
	CDH23.3	1358	0	0	0	4	113	264	606	351	20	0	0	0
	CDH26.7	394	0	0	0	1	21	66	193	113	1	0	0	0
(14) Wind	WSAvg	3.7	4.0	3.9	4.0	3.6	3.5	3.4	3.2	3.2	3.2	3.7	4.0	4.2
(15) Precipitation	PrecAvg	665	47	39	36	34	55	71	79	74	66	66	52	45
	PrecMax	878	84	78	73	57	116	118	167	145	185	134	87	80
	PrecMin	437	16	8	14	12	17	9	13	13	12	16	3	14
	PrecStd	108	17	14	16	13	25	26	40	37	33	36	19	19
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	4.0	13.9	23.8	29.1	30.9	33.9	34.0	25.9	19.9	12.1	8.0
		MCWB	3.9	2.9	8.0	13.1	18.5	19.1	20.2	19.5	16.5	13.6	10.4	7.2
	2%	DB	2.9	2.9	9.1	20.1	26.2	28.8	30.9	29.2	23.1	16.1	9.1	5.8
		MCWB	2.1	1.3	4.5	11.7	17.0	18.5	19.4	19.4	15.6	12.5	8.0	5.0
	5%	DB	1.9	2.0	6.8	17.8	24.1	26.6	28.9	27.0	21.0	13.9	7.8	3.9
		MCWB	1.2	1.0	3.3	10.5	15.4	17.8	19.2	18.6	15.4	11.2	6.8	3.1
	10%	DB	0.9	1.1	4.2	15.1	22.0	24.1	26.9	24.9	18.8	12.0	5.9	2.1
		MCWB	0.3	0.5	2.0	9.0	14.1	16.8	18.8	17.9	14.0	10.0	5.1	1.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	3.0	8.0	14.1	19.4	21.6	22.8	22.0	18.1	14.7	10.7	7.1
		MCDB	4.4	3.6	12.3	21.5	26.8	27.6	28.1	28.4	22.9	17.7	12.3	7.6
	2%	WB	1.8	1.6	5.2	12.5	17.8	20.0	21.2	20.5	16.6	13.0	8.0	4.7
		MCDB	2.3	2.0	8.0	18.1	24.3	25.7	26.9	26.4	21.0	15.6	8.7	5.2
	5%	WB	0.9	0.9	3.6	11.2	16.6	18.8	20.4	19.5	15.5	11.5	6.4	2.8
		MCDB	1.3	1.4	5.8	16.4	22.0	24.2	26.2	25.3	19.7	13.9	7.1	3.4
	10%	WB	0.0	0.4	2.3	9.9	15.4	17.6	19.6	18.5	14.4	10.2	4.7	1.3
		MCDB	0.6	1.0	4.1	14.2	20.1	22.5	25.1	23.7	18.1	11.8	5.4	1.7
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	6.5	7.6	10.3	11.7	11.1	11.0	10.7	9.2	6.2	4.4	4.4
		MCDBR	3.7	4.6	10.2	14.3	14.7	14.7	14.3	14.1	13.1	9.8	6.2	4.0
	5% WB	MCWBR	3.7	4.1	6.5	7.6	6.7	6.5	5.3	5.7	6.8	6.3	5.2	3.9
		MCDWR	3.7	3.8	8.6	12.8	13.1	12.7	12.0	12.4	11.3	8.7	5.2	3.8
(40) Clear-Sky Solar Irradiance	taub		0.291	0.332	0.350	0.424	0.401	0.389	0.434	0.440	0.391	0.360	0.325	0.291
		taud	2.111	2.049	2.102	2.101	2.290	2.345	2.242	2.207	2.325	2.340	2.232	2.122
	Ebn at Noon		600	705	801	790	841	856	804	769	761	689	563	505
		Edh at Noon	63	98	118	138	121	116	127	123	96	75	57	50
(44) All-Sky Solar Radiation	RadAvg		0.62	1.49	3.02	4.01	5.31	5.50	5.42	4.42	2.80	1.29	0.51	0.34
	RadStd		0.12	0.20	0.31	0.34	0.48	0.53	0.56	0.48	0.37	0.22	0.09	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.69	N/A	N/A	N/A	N/A	N/A	N/A	+155	N/A
(47) Regional (7 neighbors)		+0.75	N/A	N/A	N/A	N/A	N/A	-157	-224	+116

Nomenclature: See separate page